Lectures 02

Math 4741: Mathematical Analysis

Chapter 2: Introduction to Stochastic Processes

Texts: 1) Markov Processes for Stochastic Modeling – Oliver Ibe

2) Probability and Random Processes – Scott Miller

Stochastic Processes

In Computer Engineering, we frequently encounter phenomena that are random function of time:

1) Number of packets entering a server
2) Number of hits on a web page
3) Signal strength in a mobile phone

Such random phenomena are modeled by stochastic processes.

Topics covered in this lecture are:

1. Basic definition of stochastic process and examples
2. Ensemble concepts
3. Probability Models associate with stochastic processes
4. Stationary stochastic processes
5. Ergodic stochastic processes

Basic Definition of Stochastic Processes

We extend the model of Random Variables to define SP

- Random Variables: X, is a function of the possible outcome, w, of an experiment
- Maps each outcome $w \in S$ into a point on the real line
- $X(w)$ is the function and we suppress w to represent RVs

A RV X represents a mapped real value associated with $w \in S$.

And for Stochastic Process (SP),

- Extend the concept of RV by including time
- It is function of the outcome of an experiment $w \in S$ that maps each $w \in S$ to different real numbers at different times
- SP maps each w into a time function $x(t)$

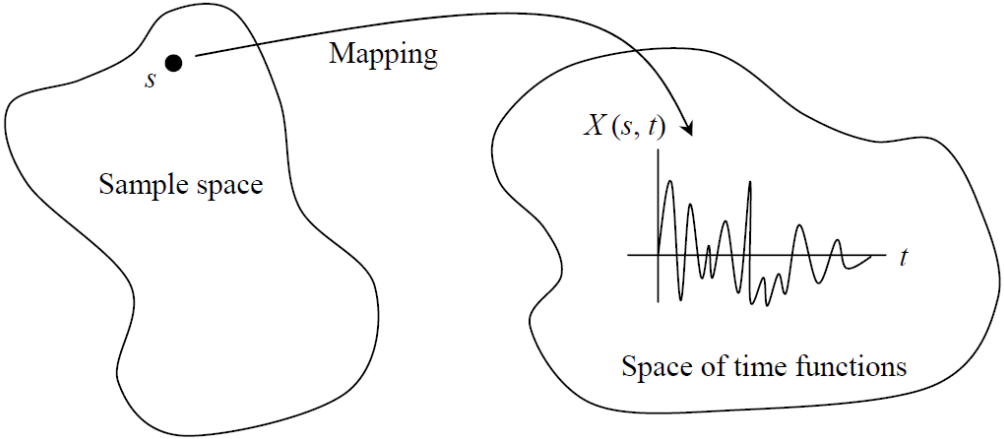
Definition:

A SP process is a function of the elements of a sample space S, as well as another independent variable, t, usually time.

Given an experiment with sample space S, an SP, $X(t, w)$, maps each possible outcome, $w \in S$, a function of time t, $x(t)$, specified by some rules.

Like the random variable, we suppress the element of the sample space w and denote the stochastic process by $\boldsymbol{X(t)}$**.**

- The word stochastic means random and process means function of time.
- A stochastic process is a mathematical model of a probabilistic experiment that evolves in time and generates a sequence of numerical values.
- The terms stochastic and random will be used interchangeably

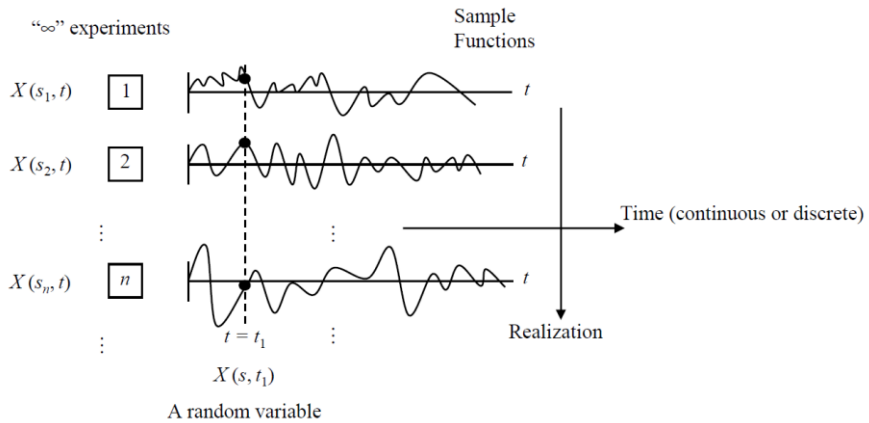


The function $x(t)$ is a member of a family/set/collection of functions

- $X(t)$ is the random process and $x(t)$ is one particular member or realization of the RP

A RP/SP has the following interpretations:

1. If we fix $t = t_1$, then $X(t_1, w)$ is a RV
2. If we fix the outcome of the experiment to w_1, then $X(t, w_1)$ is a sample realization of the SP
3. If we fix both: $t = t_1$ and $w = w_1$, then $X(t_1, w_1)$ is a real number
4. $X(t, w)$ is a collection of realizations or sample functions and is the stochastic process.



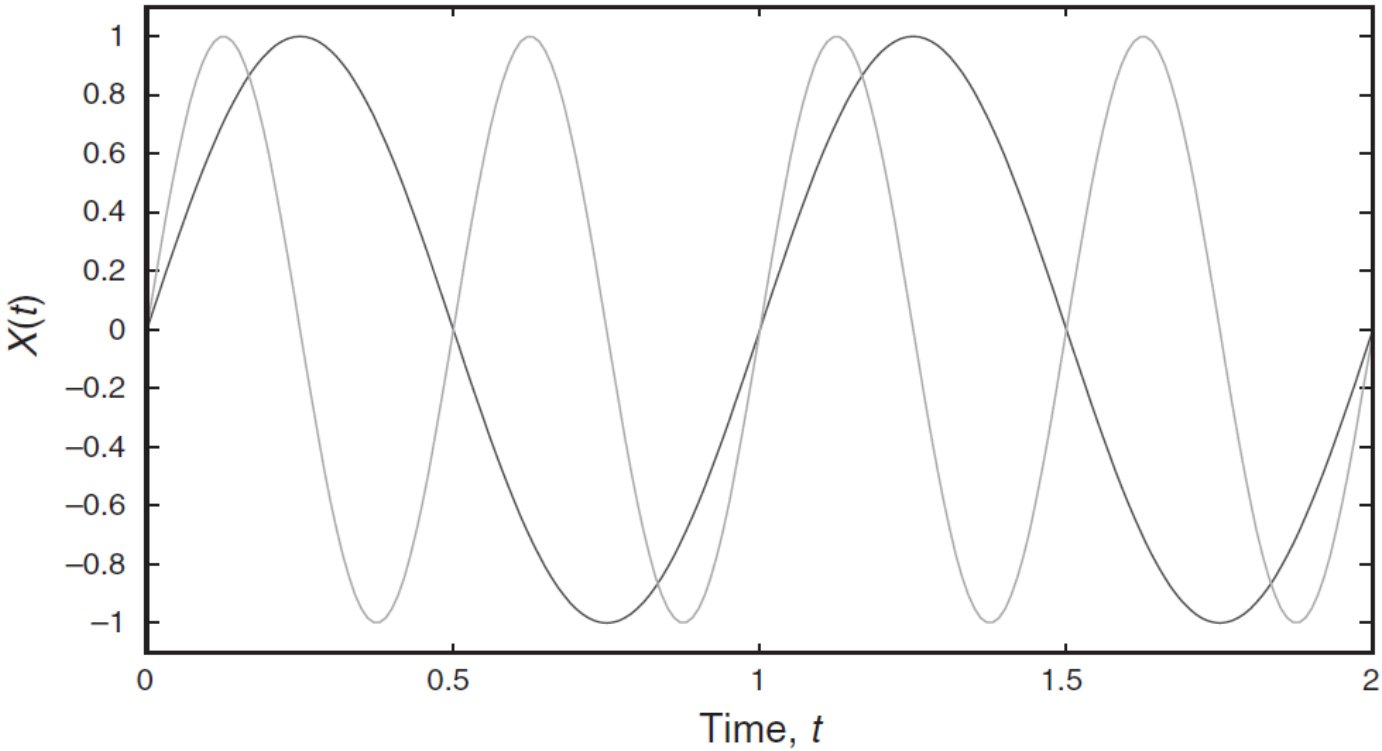
Examples of SPs

<u>Example 1</u>

Consider an experiment: Flipping a coin

If the outcome is a head, $w = H$, then $X(t)$ maps w to the function $x_H(t) = \sin(\omega_0 t)$

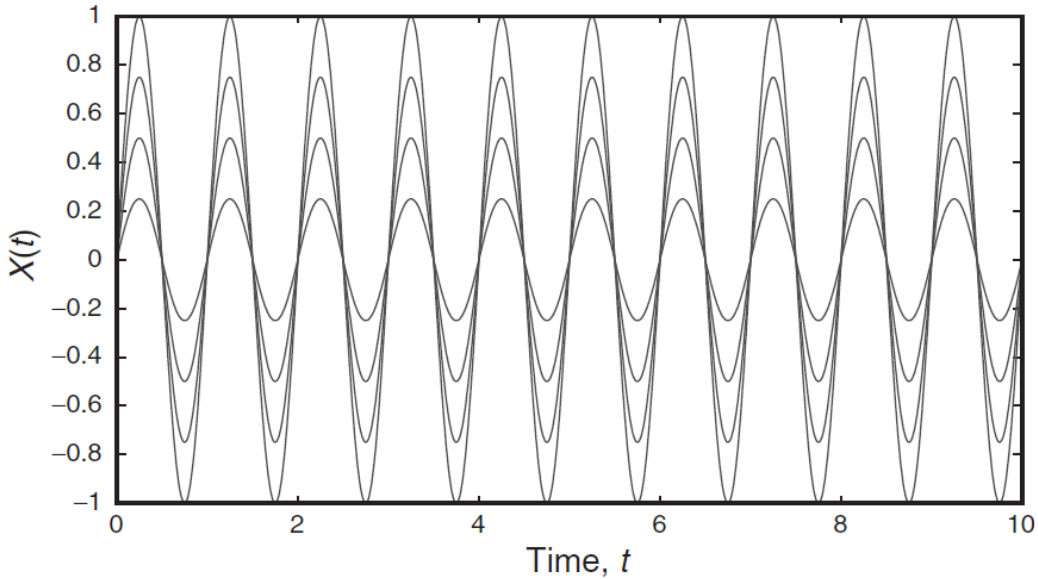
If the outcome is a tail, $w = T$, then $X(t)$ maps w to the function $x_T(t) = \sin(2\omega_0 t)$


1
0.8
0.6
0.4
0.2
0
−0.2
−0.4
−0.6
−0.8
−1
X(t)
0
0.5
1
1.5
2
Time, t

Example 2

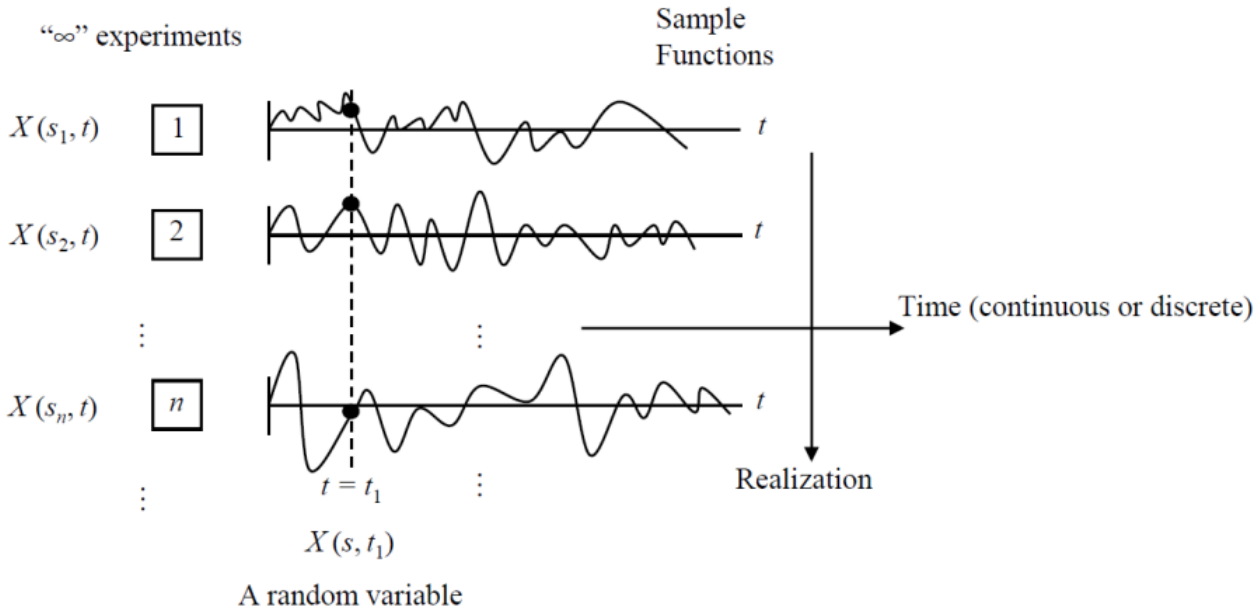
Experiment: $A \sim unif\,[0.1)$

RP: $X(t) = A\sin(\omega t)$



Ensemble Concept:

1. The family of functions is known as ensemble
2. An ensemble may consist of an infinite number of functions
3. An ensemble can be considered as having two directions
    a. Towards time
    b. Towards realization
4. Given a fixed sample point/outcome $w \in S$, gives a fixed sample function
    a. One can proceed in the time direction
5. Given a fixed time, we have a collection of samples, time samples,
    a. One can proceed along the realization
    b. Possible values of a random variable


"∞" experiments
Sample Functions
$X(s_1, t)$
1
$X(s_2, t)$
2
$X(s_n, t)$
n
t
Time (continuous or discrete)
Realization
$t = t_1$
$X(s, t_1)$
A random variable

Any time sample of an SP is a random variable

<u>Accordingly, we can consider that an SP is a collection of random variables</u>

May be infinite in number

Mathematical tool for studying RPs

Every RV can be characterized by a probability model: a CDF and a PDF or PMF

Consider a sample of $X(t)$ at $t = t_0$, then $X(t_0)$ is a RV – with possible distribution functions:

PDF: $f_X(x;\ t_1)$

CDF: $F_X(x; t_1)$

PMF: $P_X(x; t_1)$

Example 3: Telegraph signal

A transition from 0 ->1 or 1 -> 0 occurs after an independent exponential time

$T_1, T_2, T_3, \ldots$ are IID exponential random variables.

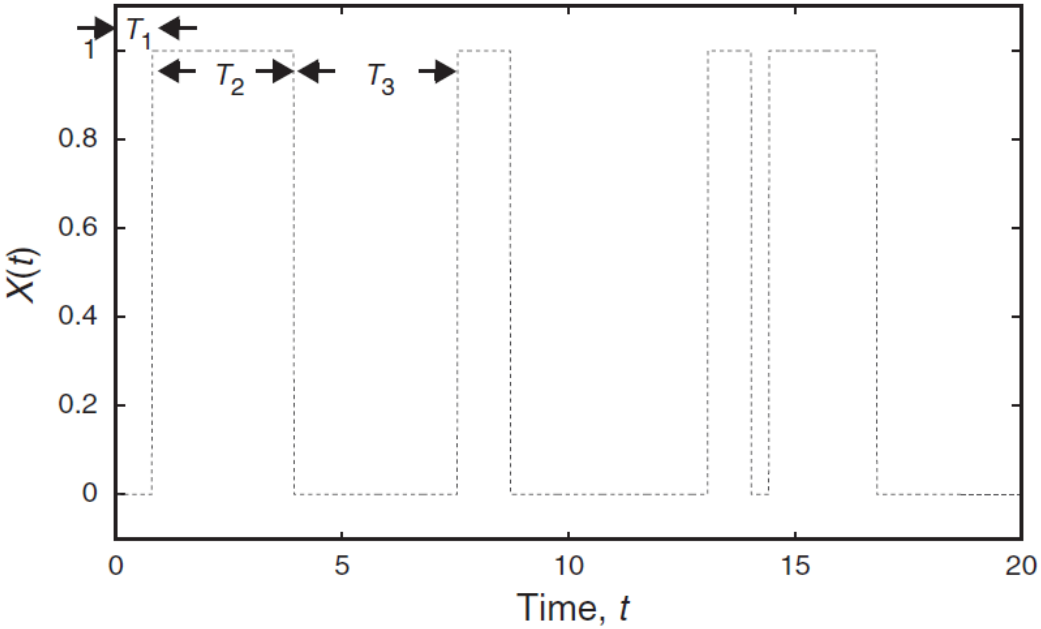


Figure 8.4 One possible realization for the random telegraph signal of Example 8.4.

The SP is binary valued, time samples are Bernoulli random variables

What is the probability that $X_k = X(t_k)$ is equal to 1 or 0?

If there are n transitions in $[0, t_k) : X(t_k) = n \; mod \; 2$

When number of transitions is odd => $X(t_k) = 1$

When number of transitions is even => $X(t_k) = 0$

$$P[X(t_k) = 0] = \frac{1}{2} + \frac{1}{2} e^{-2\lambda t_k}$$

$$P[X(t_k) = 1] = \frac{1}{2} - \frac{1}{2} e^{-2\lambda t_k}$$

For two time-samples at $t = t_1$ and $t = t_2$, the PDF $f_X(x, t)$ describe two RVs X_1 and X_2, but it does not describe relation between X_1 and X_2.

For SPs, X_1 and X_2 might be highly correlated, if t_1 and t_2 are close

while X_1 and X_2 might be uncorrelated, if t_1 and t_2 are far apart

We need to know the joint distributions – second order PDF, CDF and PMF

Second-order Distribution of an SP

Two samples at $t = t_1$ and $t = t_2$

JPDF: $f_{X_1X_2}(x_1, x_2; t_1, t_2)$

JCDF: $F_{X_1X_2}(x_1, x_2; t_1, t_2)$

JPMF: $P_{X_1X_2}(x_1, x_2; t_1, t_2)$

Can be extended for any number of samples: Multivariate distributions

Mean Function:

DEFINITION 8.2: The *mean function* of a random process is simply the expected value of the process. For continuous time processes this is written as

$$\mu_X(t) = E[X(t)] = \int_{-\infty}^{\infty} x f_X(x;t)\,dx, \tag{8.1}$$

while for discrete time processes, the following notation is used:

$$\mu_X[n] = E[X[n]] = \int_{-\infty}^{\infty} x f_X(x;n)\,dx. \tag{8.2}$$

EXAMPLE 8.8: Consider the random telegraph process of Example 8.4. It was shown in Example 8.6 that the first order PMF of this process was described by a Bernoulli distribution with

$$\Pr(X(t) = 1) = \tfrac{1}{2} - \tfrac{1}{2}\exp(-\lambda t).$$

The mean function then follows as

$$\mu_X(t) = E[X(t)] = 1^* \Pr(X(t) = 1) + 0^* \Pr(X(t) = 0) = \tfrac{1}{2} - \tfrac{1}{2}\exp(-\lambda t).$$

EXAMPLE 8.9: Next, consider the sinusoidal random process of Example 8.2 where $X(t) = A\sin(\omega_0 t)$ and A was a uniform random variable over [0, 1). In this case,

$$\mu_X(t) = E[X(t)] = E[A\sin(\omega_0 t)] = E[A]\sin(\omega_0 t) = \frac{1}{2}\sin(\omega_0 t).$$

This example illustrates a very important concept in that quite often it is not necessary to explicitly evaluate the first order PDF of a random process in order to evaluate its mean function.

EXAMPLE 8.10: Now suppose the random process of the previous example is slightly modified. In particular, consider a sine-wave process where the random variable is the phase, Θ, which is uniformly distributed over $[0, 2\pi)$. That is, $X(t) = a\sin(\omega_0 t + \Theta)$. For this example, the amplitude of the sine wave, a, is taken to be fixed (not random). The mean function is then

$$\mu_X(t) = E[X(t)] = E[a\sin(\omega_0 t + \Theta)] = a\int f_\Theta(\theta)\sin(\omega_0 t + \theta)\,d\theta$$

$$= \frac{a}{2\pi}\int_0^{2\pi} \sin(\omega_0 t + \theta)\,d\theta = 0,$$

which is a constant. Why is the mean function of the previous example a function of time and this one is not? Consider the member functions of the respective ensembles for the two random processes.

Time average of a stochastic Process

Average taken in time over a single realization of the RP, known as the time averaging or temporal averaging

$$\langle x(t) \rangle = \lim_{T \to \infty} \frac{1}{2T} \int_{-T}^{T} x(t) dt$$

DEFINITION 8.3: The *autocorrelation function*, $R_{XX}(t_1, t_2)$, of a continuous time random process, $X(t)$, is defined as the expected value of the product $X(t_1)X(t_2)$:

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = \iint_{-\infty}^{\infty} x_1 x_2 f_{X_1,X_2}(x_1, x_2; t_1, t_2)\, dx_1\, dx_2. \tag{8.3}$$

For discrete time processes, the autocorrelation function is

$$R_{XX}[n_1, n_2] = E[X[n_1]X[n_2]] = \iint_{-\infty}^{\infty} x_1 x_2 f_{X_1,X_2}(x_1, x_2; n_1, n_2)\, dx_1\, dx_2. \tag{8.4}$$

EXAMPLE 8.12: Consider the sine wave process with a uniformly distributed amplitude as described in Examples 8.2 and 8.9, where $X(t) = A\sin(\omega_o t)$. The autocorrelation function is found as

$$\begin{aligned} R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] &= E[A^2 \sin(\omega_o t_1)\sin(\omega_o t_2)] \\ &= \tfrac{1}{3}\sin(\omega_o t_1)\sin(\omega_o t_2), \end{aligned}$$

or

$$R_{XX}(t, t+\tau) = \tfrac{1}{3}\sin(\omega_o t)\sin(\omega_o(t+\tau)).$$

EXAMPLE 8.13: Now consider the sine wave process with random phase of Example 8.10 where $X(t) = a\sin(\omega_0 t + \Theta)$. Then

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)] = E[a^2 \sin(\omega_0 t_1 + \theta)\sin(\omega_0 t_2 + \theta)].$$

To aid in calculating this expected value, we use the trigonometric identity

$$\sin(x)\sin(y) = \tfrac{1}{2}\cos(x - y) - \tfrac{1}{2}\cos(x + y).$$

The autocorrelation then simplifies to

$$\begin{aligned} R_{XX}(t_1, t_2) &= \frac{a^2}{2}E[\cos(\omega_0(t_2 - t_1))] + \frac{a^2}{2}E[\cos(\omega_0(t_1 + t_2 + 2\theta))] \\ &= \frac{a^2}{2}\cos(\omega_0(t_2 - t_1)), \end{aligned}$$

or

$$R_{XX}(t, t + \tau) = \frac{a^2}{2}\cos(\omega_0 \tau).$$

DEFINITION 8.4: The autocovariance function, $C_{XX}(t_1, t_2)$, of a continuous time random process, $X(t)$, is defined as the covariance of $X(t_1)$ and $X(t_2)$:

$$C_{XX}(t_1, t_2) = \text{Cov}(X(t_1), X(t_2)) = E[(X(t_1) - \mu_X(t_1))(X(t_2) - \mu_X(t_2))]. \tag{8.6}$$

The definition is easily extended to discrete time random processes.

As with the covariance function for random variables, the autocovariance function can be written in terms of the autocorrelation function and the mean function:

$$C_{XX}(t_1, t_2) = R_{XX}(t_1, t_2) - \mu_X(t_1)\mu_X(t_2). \tag{8.7}$$

DEFINITION 8.5: For a pair of random processes $X(t)$ and $Y(t)$, the crosscorrelation function is defined as

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]. \tag{8.10}$$

Likewise, the cross-covariance function is

$$C_{XY}(t_1, t_2) = E[(X(t_1) - \mu_X(t_1))(Y(t_2) - \mu_Y(t_2))]. \tag{8.11}$$

EXAMPLE 8.15: Suppose $X(t)$ is a zero-mean random process with autocorrelation function $R_{XX}(t_1, t_2)$. A new process $Y(t)$ is formed by delaying $X(t)$ by some amount t_d. That is, $Y(t) = X(t - t_d)$. Then the crosscorrelation function is

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)] = E[X(t_1)X(t_2 - t_d)] = R_{XX}(t_1, t_2 - t_d).$$

In a similar fashion, it is seen that $R_{YX}(t_1, t_2) = R_{XX}(t_1 - t_d, t_2)$ and $R_{YY}(t_1, t_2) = R_{XX}(t_1 - t_d, t_2 - t_d)$.

Stationary RPS

Mean, autocorrelation and autocovariance can provide information about the temporal structure of a random process.

DEFINITION 8.6: A continuous time random process $X(t)$ is *strict sense stationary* if the statistics of the process are invariant to a time shift. Specifically, for any time shift τ and any integer $n \geq 1$,

$$f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n; t_1, t_2, \ldots, t_n)$$
$$= f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n; t_1 + \tau, t_2 + \tau, \ldots, t_n + \tau). \qquad (8.12)$$

<u>The PDF and n-th order PDF are independent of the starting time – does not depend where it starts.</u>

show only that one PDF of any order is not invariant to a time shift. One example of a process that can be shown to be stationary in the strict sense is an IID process. That is, suppose $X(t)$ is a random process that has the property that $X(t)$ has an identical distribution for any t and that $X(t_1)$ and $X(t_2)$ are independent for any $t_1 \neq t_2$. In this case,

$$f_{X_1,X_2,\ldots,X_n}(x_1, x_2, \ldots, x_n; t_1, t_2, \ldots, t_n) = \prod_{i=1}^{n} f_{X_i}(x_i; t_i). \tag{8.13}$$

Since the nth order PDF is the product of first order PDFs and the first order PDF is invariant to a time shift, then the nth order PDF must be invariant to a time shift.

EXAMPLE 8.16: Consider the sinusoidal process with random amplitude from Example 8.2, where $X(t) = A\sin(\omega_0 t)$. This process is clearly not stationary since if we take any realization of the process, $x(t) = a\sin(\omega_0 t)$, then a time shift $x(t+\tau) = a\sin(\omega_0(t+\tau))$ would not be a realization in the original ensemble. Now suppose the process has a random phase rather than a random amplitude as in Example 8.10, resulting in $X(t) = a\sin(\omega_0 t + \Theta)$. It was already shown in Example 8.10 that $\mu_X(t) = 0$ for this process, and hence the mean function is invariant to a time shift. Furthermore, in Example 8.13 it was shown that $R_{XX}(t, t+\tau) = (a^2/2)\cos(\omega_0\tau)$, and hence the autocorrelation function is also invariant to a time shift. It is not difficult to show that the first order PDF follows an arcsine distribution:

$$f_X(x;t) = \frac{1}{\pi\sqrt{1-x^2}}, \quad -1 < x < 1,$$

DEFINITION 8.7: A random process is *wide sense stationary* (WSS) if the mean function and autocorrelation function are invariant to a time shift. In particular, this implies that

$$\mu_X(t) = \mu_X = \text{constant}, \tag{8.14}$$

$$R_{XX}(t, t+\tau) = R_{XX}(\tau) \quad \text{(function only of } \tau\text{)}. \tag{8.15}$$

EXAMPLE 8.17: Suppose we form a random process $Y(t)$ by modulating a carrier with another random process, $X(t)$. That is, let $Y(t) = X(t)\cos(\omega_0 t + \Theta)$ where Θ is uniformly distributed over $[0, 2\pi)$ and independent of $X(t)$. Under what conditions is $Y(t)$ WSS? To answer this, we calculate the mean and autocorrelation function of $Y(t)$.

$$\mu_Y(t) = E[X(t)\cos(\omega_0 t + \Theta)] = E[X(t)]E[\cos(\omega_0 t + \Theta)] = 0;$$

$$\begin{aligned}
R_{YY}(t,t+\tau) &= E[X(t)X(t+\tau)\cos(\omega_0 t)\cos(\omega_0(t+\tau))] \\
&= E[X(t)X(t+\tau)]\left\{\tfrac{1}{2}\cos(\omega_0\tau) + \tfrac{1}{2}E[\cos(\omega_0(2t+\tau)+2\Theta)]\right\} \\
&= \tfrac{1}{2}R_{XX}(t,t+\tau)\cos(\omega_0\tau)
\end{aligned}$$

$$\langle x(t)\rangle = \lim_{t_o\to\infty} \frac{1}{2t_o}\int_{-t_o}^{t_o} x(t)\,dt.$$

ergodic in the mean.

$$\langle x(t)x(t+\tau)\rangle = \lim_{t_0\to\infty} \frac{1}{2t_0}\int_{-t_0}^{t_0} x(t)x(t+\tau)\,dt.$$

ergodic in the autocorrelation

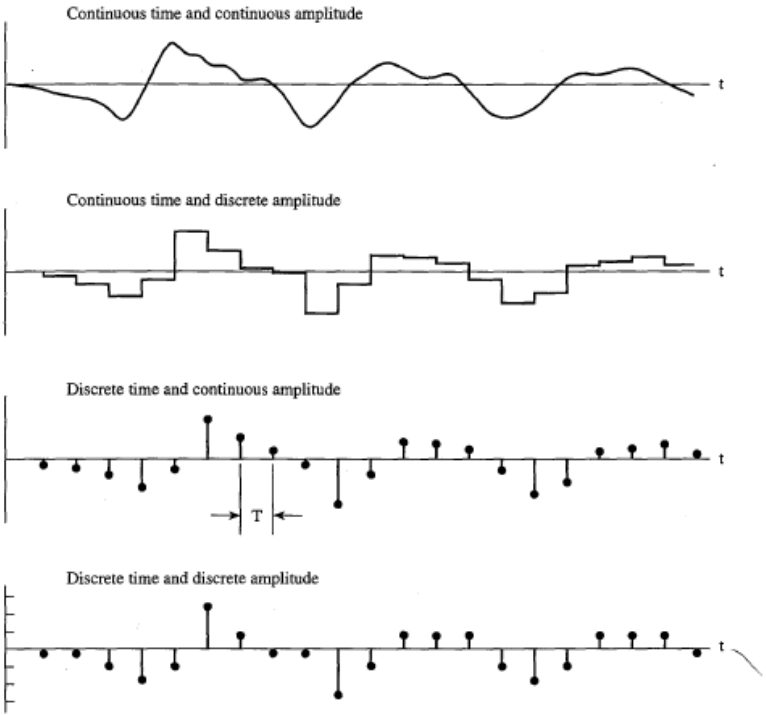


Figure 5.1 Examples of four types of signals